

- Concise Functional Requirements Document (FRD)**
- Introduction:**

The purpose of this FRD is to outline the functional requirements for the "Application", a data integration and reporting solution that consolidates customer order data from CSV files into a structured database.

- Functional Requirements:**

1. **Data Ingestion Module**

- Read CSV data from specified locations and load it into Delta tables "customer" and "order".
- Handle exceptions for file not found or invalid file format.

2. **Data Transformation Module**

- Remove "Null"/null and duplicate records from "customer" and "order" tables.
- Validate data types and schema of ingested data.

3. **Data Integration Module**

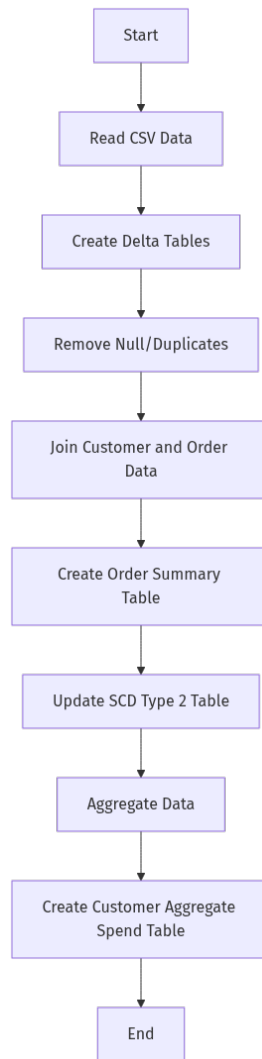
- Create "ordersummary" table if it does not exist and load joined data into an SCD Type 2 table.
- Update SCD Type 2 "ordersummary" table whenever there is a change in the "customer" table.

4. **Data Aggregation Module**

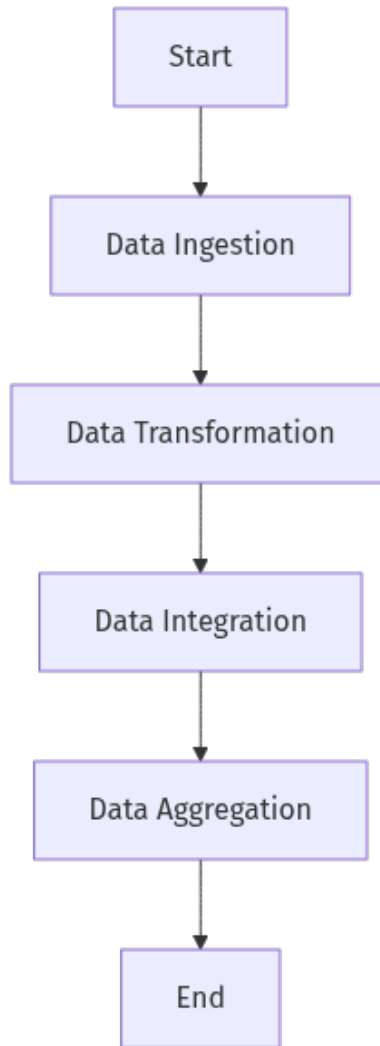
- Create "customeraggregatespend" table if it does not exist and load aggregated data into it.
- Aggregate "TotalAmount" column from "ordersummary" table and group it by "Name" and "Date" columns.

- Detailed Functional Requirements:**

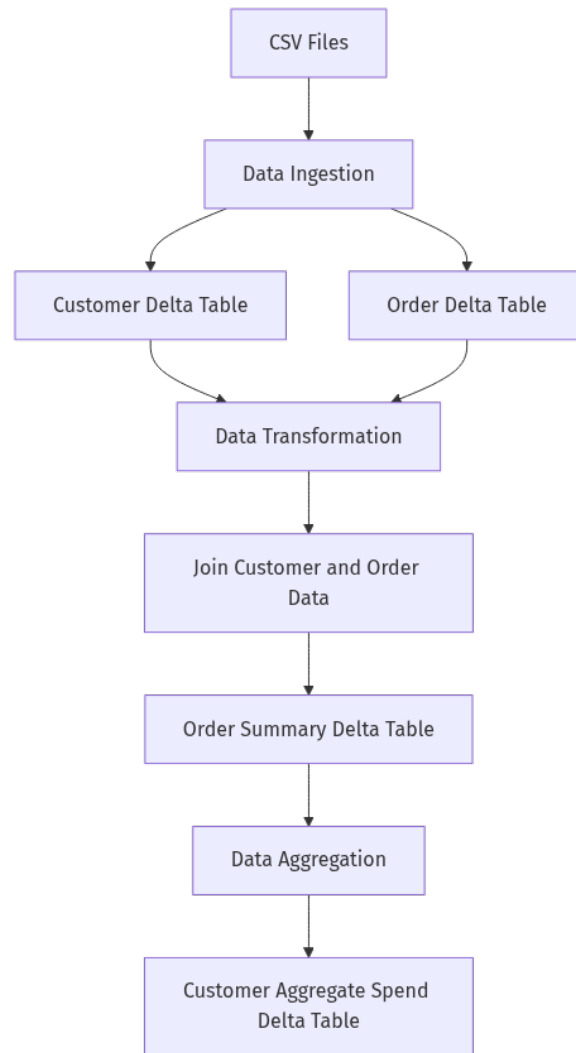
- The system should be able to read CSV data from
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.
- The system should be able to create Delta tables "customer" and "order" with the correct schema.
- The system should be able to join "customer" and "order" data using the "CustId" field.
- The system should be able to update the SCD Type 2 "ordersummary" table by making old records inactive and new records active, and updating `StartDate` and `EndDate` accordingly.



- System Flow Diagram**



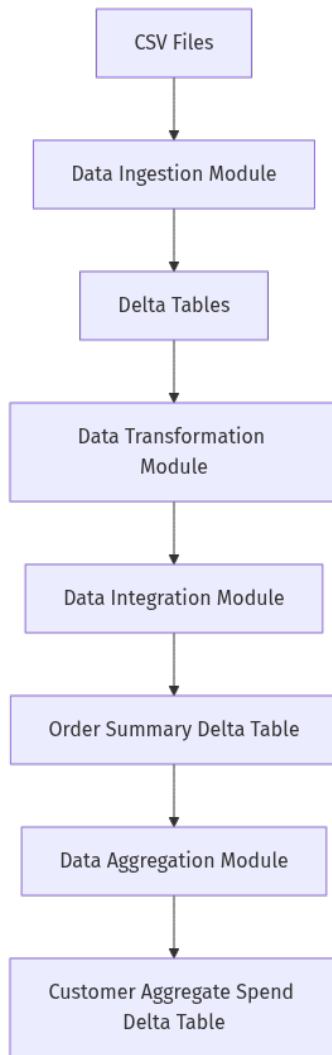
- Process Flow / Flowchart**



- Data Flow Diagram (DFD)**

[Diagram failed: Kroki render failed (HTTP 400): Error 400: SyntaxError: Parse error on line 2: ...aph TD;participant Application as "Appl -----^ Expecting 'SEMI', 'NEWLINE', 'EOF', 'AMP', 'START_LINK', 'LINK', 'LINK_ID', got 'NODE_STRING' Error: Syntax error in graph at Worker.convert (file:///usr/local/kroki/src/worker.js:44:15) at async file:///usr/local/kroki/src/index.js:31:28]

- Sequence Diagram**



- High-Level Architecture Diagram**